

SolaX C&I Storage

Fire Strategy Elements



Fire Protection Design and Layout Optimization:

1. **Independent Fire Partitions:** The energy storage system is separated from other facilities, with installation distances meeting local regulatory requirements to reduce the risk of fire spread.
2. **Ventilation System:** A well-designed ventilation system prevents the accumulation of harmful gases, ensuring system safety.
3. **Fire Walls and Fire-Resistant Materials:** Fire walls are installed according to local regulations, and SolaX cabinets use UL94V0-rated fire-resistant materials, further enhancing system safety.

Monitoring and Early Warning System:

1. **Temperature Sensors:** Real-time detection of ambient temperatures, with adjustments to internal cabinet temperatures based on environmental temperature changes, achieving cell-level monitoring and early warning.
2. **Smoke Detectors:** Real-time detection of smoke signals to ensure system safety.
3. **Gas Sensors:** Real-time detection of harmful gases, triggering the ventilation system to prevent accumulation.
4. **Remote Monitoring System:** Real-time monitoring of system operation status through EMS for timely handling of anomalies.

Fire Suppression System:

1. **Gas Fire Suppression System:** The energy storage system is equipped with a gas aerosol fire unit capable of heat absorption cooling, chemical suppression, and oxygen isolation.
2. **Sprinkler System:** local fire hydrants for water supply, with fire sprinkler units installed on top of the cabinets, providing comprehensive protection with dual fire protection design including gas aerosol and water firefighting.

Exhaust and Explosion Prevention:

1. **Safety Exhaust Ports:** Designed pressure relief ports on the energy storage system enclosures or cabinets to reduce internal pressure and ensure system safety.
2. **Explosion-Proof Ventilation System:** When the system detects combustible gases exceeding the threshold, it uses explosion-proof fans to expel harmful gases, reducing the concentration of combustible gases inside the cabinet and ensuring system safety.

Emergency Response and Isolation Strategy:

1. **Emergency Power Cut:** In case of anomalies in the energy storage system, the system automatically cuts off the power supply from both AC and DC sides, triggers a local alarm, and uploads the alarm information to the cloud platform.
2. **Fire Isolation Strategy:** Containing system anomalies within a single cabinet to prevent affecting the safety of adjacent equipment, with a compartmentalized design that can withstand fire hazards for 1.5 hours.

These strategies together form a comprehensive and multi-layered safety protection system aimed at maximizing the protection of personnel safety and the stable operation of equipment.